

LAST NAME: _____

Practice Quiz 05A — MATH 2055 — Fall 2026

Practice for Quiz 05

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

Recall that $d \mid n$ means $n = dk$ for some integer k .

1. Prove

$$\forall x \in \mathbf{Z} \left(4 \nmid x^2 \longrightarrow 2 \nmid x \right).$$

Introduce the arbitrary integer explicitly and name your proof strategy.